



The Normal Distribution Topic Test 1 Solution

Total Marks: 30

Time: 50 minutes

- 1 Remy achieved a z-score of 2.2 on her latest Geography test. If the test scores were normally distributed with a mean of 64 and a standard deviation of 5, what was Remy's actual test mark?

$$z = \frac{x - \bar{x}}{\sigma}$$

1

$$2.2 = \frac{x - 64}{5}$$

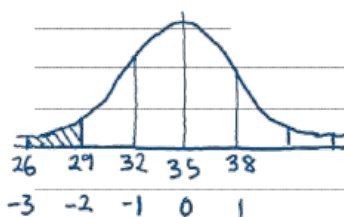
$$11 = x - 64$$

$$75 = x$$

$$x = 75$$

- 2 In a normally distributed set of scores, the mean is 35 and the standard deviation is 3. What percentage of scores lie between 26 and 29?

2

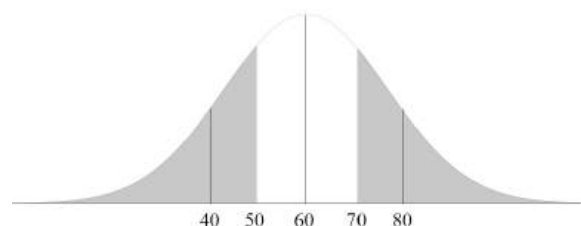


$$\frac{99.7}{2} - \frac{95}{2} = 2.35$$

$\therefore 2.35\%$ of scores lie between 26 and 29.

- 3 The normal distribution shows the results of a Mathematics Standard assessment task. It has a mean of 60 and a standard deviation of 10. What percentage of results lies in the shaded region?

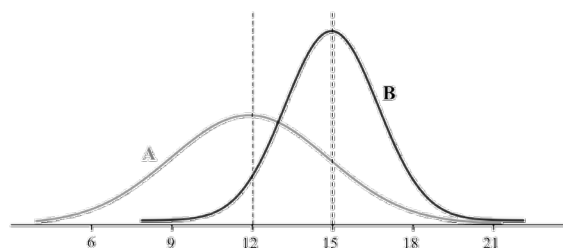
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$$100\% - 68\% = 32\%$$

- 4 The diagram below shows two different normal distribution curves A and B. Compare the curves using the terms 'mean' and 'standard deviation'.

2



Curve A has a lower mean but higher standard deviation than curve B.

- 5 After completing a numeracy test (%), Hanna had achieved a z-score of -2.

- a. How does Hanna's result compare with those of her peers? Explain your answer in terms of the mean and standard deviation. Hazel's score is 2 standard deviation below the mean. 1
- b. If the mean test mark was 86 and the standard deviation was 4, calculate Hanna's actual mark. 2

$$z = \frac{x - \mu}{\sigma}$$

$$-2 = \frac{x - 86}{4}$$

$$x - 86 = -8$$

$$x = 78$$

Hanna's actual mark was 78.

- 6 At a school carnival, the final heights jumped by year 12 competitors were found to be normally distributed with a mean of 5.54 metres and a standard deviation of 3 metres. The standard normal distribution table is shown. The values in the table represent the area under the normal curve to the left of the z-score. 3

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.50000	0.50399	0.50798	0.51197	0.51595	0.51994	0.52392	0.52790	0.53188	0.53586
0.1	0.53983	0.54380	0.54776	0.55172	0.55567	0.55966	0.56360	0.56749	0.57142	0.57535
0.2	0.57926	0.58317	0.58706	0.59095	0.59483	0.59871	0.60257	0.60642	0.61026	0.61409
0.3	0.61791	0.62172	0.62552	0.62930	0.63307	0.63683	0.64058	0.64431	0.64803	0.65173
0.4	0.65542	0.65910	0.66276	0.66640	0.67003	0.67364	0.67724	0.68082	0.68439	0.68793
0.5	0.69146	0.69497	0.69847	0.70194	0.70540	0.70884	0.71126	0.71566	0.71904	0.72240
0.6	0.72575	0.72907	0.73237	0.73565	0.73891	0.74215	0.74537	0.74857	0.75175	0.75490
0.7	0.75804	0.76115	0.76424	0.76730	0.77035	0.77337	0.77637	0.77935	0.78230	0.78524
0.8	0.78814	0.79103	0.79389	0.79673	0.79955	0.80234	0.80511	0.80785	0.81057	0.81327
0.9	0.81594	0.81859	0.82121	0.82381	0.82639	0.82894	0.83147	0.83398	0.83646	0.83891
1.0	0.84134	0.84375	0.84614	0.84849	0.85083	0.85314	0.85543	0.85769	0.85993	0.86214

Using standardised scores and the standard normal distribution table, find the probability that a randomly chosen competitor achieves height between 5.99 and 6.59 metres. Give your answer correct to three decimal places.

$$z = \frac{x - \mu}{\sigma}$$

For $x = 5.99$

For $x = 6.59$

$$z = \frac{5.99 - 5.54}{3} = 0.15$$

$$z = \frac{6.59 - 5.54}{3} = 0.35$$

$$P(0.15 < z < 0.35) = P(z < 0.35) - P(z < 0.15) \\ = 0.63683 - 0.55966 = 0.07717 \approx 0.077$$

- 7 The assessment marks for a Biology assessment and a History assessment are both normally distributed. The Biology assessment has a mean of 74 and a standard deviation of 3. The History assessment has a mean of 82 and a standard deviation of 1.5.

- a. Kathryn knows that her z-score for both her Biology assessment and History assessment are the same. She received a mark of 80 in Biology. Calculate her assessment mark for her history assessment. 2

$$\text{z-score (Biology)} = \frac{80 - 74}{3} = 2$$

$$\text{z-score (History)} = 2 = \frac{x - 82}{1.5}$$

$$\text{So } x - 82 = 3 \quad x = 85$$

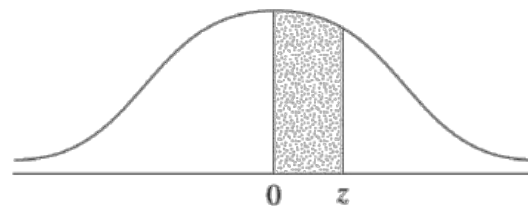
- b. What percentage of students received a lower score than Kathryn in her Biology assessment? 1

97.5%

- 8 A random variable is normally distributed with mean 0 and standard deviation 1. The table gives the probability that this random variable lies between 0 and z for different values of z . 4

z	0	0.25	0.5	0.75	1	1.25	1.5	1.75	2
Probability	0	0.0987	0.1915	0.2734	0.3413	0.3944	0.4332	0.4599	0.4773

The probability values given in the table for different values of z are represented by the shaded area in the following diagram. Tennis balls are manufactured, and their weights are normally distributed with the following specifications:



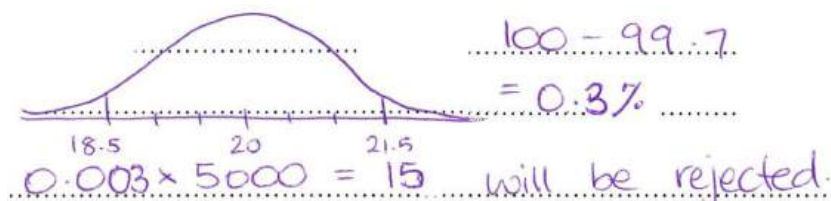
- Mean (μ) = 57.4 g
- Standard deviation (σ) = 0.4 g.

Tennis balls that weight less than 56.7 g do not meet the rules of tennis and are rejected.

Two recently manufactured tennis balls are chosen at random and weighed. By drawing a tree diagram, or otherwise, find the probability that neither tennis ball is rejected, correct to the nearest percentage.

$$\begin{aligned}\text{Min } z\text{-score} &= \frac{56.7 - 57.4}{0.4} = -1.75 \\ \text{Probability that a ball is NOT rejected:} \\ P(z > -1.75) &= P(z < 1.75) \\ &= 0.4599 + 0.5 = 0.9599 \\ \text{Probability that neither ball is rejected is} \\ 0.9599 \times 0.9599 &= 0.9214 \dots = 92\%\end{aligned}$$

- 9 A manufacturing plant produces steel balls with masses that are normally distributed. The mean mass of the balls is 20 g with a standard deviation of 0.5 g. In quality control testing of the manufactured balls, any ball with a mass greater than 21.5 g and less than 18.5 g is rejected. From a sample of 5000 steel balls tested, how many would be expected to be rejected? 2



- 10 Zane achieved a mark of 68/80 for his Biology Assessment task 3 exam. The scores for this exam are normally distributed, with a standard deviation of 3.2% and a mean of 80%. If he wants to perform at the same level in the HSC Trial exam, and the mean increases by 5% and the standard deviation decreases by 2%, what mark must he achieve, to the nearest whole mark, if the exam is out of 70? 3

$$\begin{aligned}\frac{68}{80} &= 85\% \\ z &= \frac{85 - 80}{3.2} = 1.5625 \\ 1.5625 &= \frac{x - 85}{1.2} \\ 1.875 &= x - 85 \\ x &= 86.875 \\ \frac{y}{70} \times 100 &= 86.875 \\ y &= 60.81 \approx \frac{61}{70}\end{aligned}$$

- 11 The marks in a class test are normally distributed. The mean is 100 and the standard deviation is 10.

a. Jason's mark is 115. What is his z-score?

$$z = \frac{115 - 100}{10} = 1.5$$

1

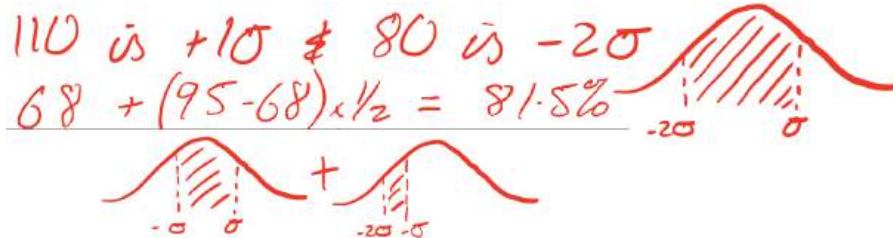
b. Mary has a z-score of 0. What mark did she achieve in the test?

1

100 (i.e. the mean)

c. What percentage of marks lie between 80 and 110?

2



- 12 The data for a large population is normally distributed. Which of the following statements is correct?

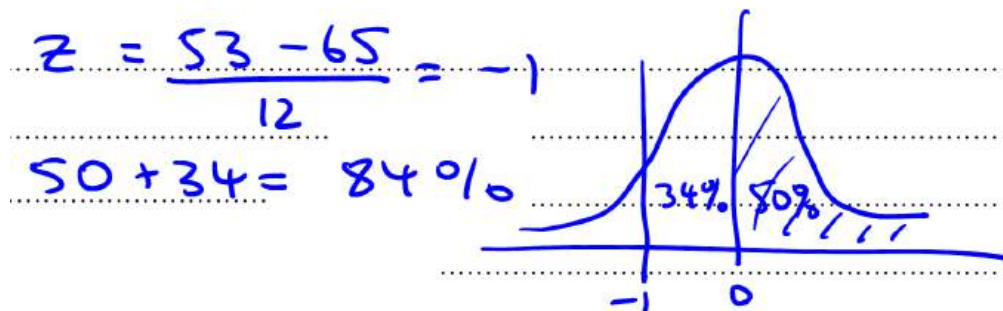
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- A. The mean, the mode and the median are all approximately equal.
- B. The mean and the mode are approximately equal but the median could be quite different.
- C. The mean and the median are approximately equal but the mode could be quite different.
- D. The mean, the mode and the median could all be quite different.

(A)

- 13 Tim scored 53 in a test for which the scores were normally distributed with a mean of 65 and a standard deviation of 12. What percentage of students scored higher than Tim?

1





The Normal Distribution Topic Test 2

Solution

Total Marks: 30

Time: 50 minutes

- 1 The table shows the mean and standard deviation of the results in Mathematics and English exams.

Subject	Mean	Standard Deviation
Mathematics	70	4
English	64	5

- a. Kate scored 78 in Mathematics and 71 in English. Calculate Kate's z-score for her results in the Mathematics and English exams. 2

For Mathematics

$$z = \frac{x - \bar{x}}{s} = \frac{78 - 70}{4} = 2$$

For English

$$z = \frac{x - \bar{x}}{s} = \frac{71 - 64}{5} = 1.4$$

- b. In which of the two exams was her result better? Justify your answer. 1

Kate scored better in Mathematics as she had a higher z score

- 2 A company claims its fluorescent lights last for 3130 hours. A sample of bulbs is tested for the operating life. The data is found to be normally distributed, with a mean of 2950 and a standard deviation of 90.

- a. What percentage of the bulbs would be expected to meet or better the claimed life of 3130 hours? 2

$$z = (3130 - 2950)/90 = 2$$

95% of scores from $z = -2$ to $z = 2$

2.5% will be above 3130 hours.

- b. The company is satisfied to have 84% of bulbs meeting or bettering its claimed life. What should that claimed life be? 2

68% between a z score of -1 and 1. Thus 32% outside this z score.

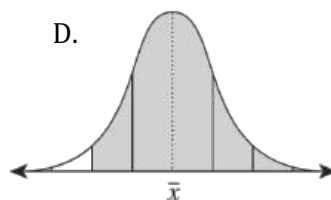
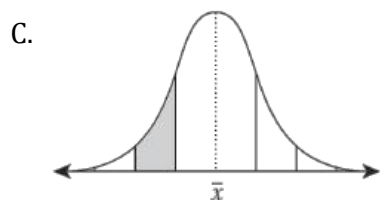
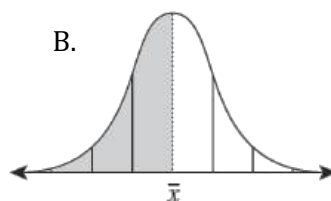
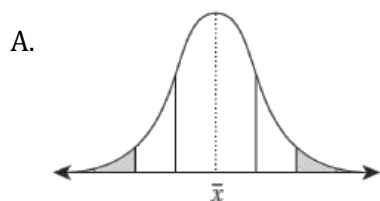
This means 16% below a z score of -1.

Thus 84% above a z score above -1.

$$-1 = (x - 2950)/90 \quad x = 2860$$

Claimed life should be 2860 hours

- 3 For a normally distributed dataset, which of the following graphs could represent the percentage of scores that have a z-score greater than -2? 1



D is correct.

- 4 Angie scored 68 in an examination where the mean was 58.4 and the standard deviation was 4.8. Amber scored 75 in an examination where the mean was 66.5 and the standard deviation was 8.5.

a. Find each girl's z-score. Angie: $z = \frac{68-58.4}{4.8} = 2$ Amber: $z = \frac{75-66.5}{8.5} = 1$ 2

- b. Which girl scored higher in comparison with the other students in their examination? 1

Angie. Her z-score of 2 is better than Amber's z-score of 1.

- c. If 200 students sat Amber's examination, how many students scored below Amber's mark? 2

approximately 68% of scores have z-scores between -1 and 1

$$64\% \div 2 = 34\%$$

$$50\% + 34\% = 84\%$$

$$84\% \times 200 = 168$$

$\therefore$ Amber did better than 168 student who did the same examination.

- 5 At a schools swimming carnival, the finish times for students under 17 who competed in the 3

50 m backstroke race are found to be normally distributed with a mean race time of 42 seconds

and a standard deviation of 3 seconds. The standard normal distribution table is shown. The

values in the table represent the area under the normal curve to the left of the z-score.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.50000	0.50399	0.50798	0.51197	0.51595	0.51994	0.52392	0.52790	0.53188	0.53586
0.1	0.53983	0.54380	0.54776	0.55172	0.55567	0.55966	0.56360	0.56749	0.57142	0.57535
0.2	0.57926	0.58317	0.58706	0.59095	0.59483	0.59871	0.60257	0.60642	0.61026	0.61409
0.3	0.61791	0.62172	0.62552	0.62930	0.63307	0.63683	0.64058	0.64431	0.64803	0.65173
0.4	0.65542	0.65910	0.66276	0.66640	0.67003	0.67364	0.67724	0.68082	0.68439	0.68793
0.5	0.69146	0.69497	0.69847	0.70194	0.70540	0.70884	0.71126	0.71566	0.71904	0.72240
0.6	0.72575	0.72907	0.73237	0.73565	0.73891	0.74215	0.74537	0.74857	0.75175	0.75490
0.7	0.75804	0.76115	0.76424	0.76730	0.77035	0.77337	0.77637	0.77935	0.78230	0.78524
0.8	0.78814	0.79103	0.79389	0.79673	0.79955	0.80234	0.80511	0.80785	0.81057	0.81327
0.9	0.81594	0.81859	0.82121	0.82381	0.82639	0.82894	0.83147	0.83398	0.83646	0.83891
1.0	0.84134	0.84375	0.84614	0.84849	0.85083	0.85314	0.85543	0.85769	0.85993	0.86214
1.1	0.86433	0.86650	0.86864	0.87076	0.87286	0.87493	0.87698	0.87900	0.88100	0.88298
1.2	0.88493	0.88686	0.88877	0.89065	0.89251	0.89435	0.89617	0.89796	0.89973	0.90147
1.3	0.90320	0.90490	0.90658	0.90824	0.90988	0.91149	0.91308	0.91466	0.91621	0.91774
1.4	0.91924	0.92073	0.92220	0.92364	0.92507	0.92647	0.92785	0.92922	0.93056	0.93189
1.5	0.93319	0.93448	0.93574	0.93699	0.93822	0.93943	0.94062	0.94179	0.94295	0.94408
1.6	0.94520	0.94630	0.94738	0.94845	0.94950	0.95053	0.95154	0.95254	0.95352	0.95449

Using standardised scores and the standard normal distribution table, find the probability that

a randomly chosen competitor achieves a finish time between 43 and 46 seconds. Give your

answer correct to three decimal places.

$$z = \frac{x - \bar{x}}{s}, \text{ where } \bar{x} = 42 \text{ and } s = 3.$$

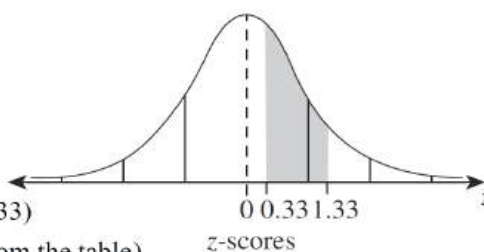
$$\text{When } x = 43, z = \frac{43 - 42}{3} = 0.3333 \dots$$

$$\text{When } x = 46, z = \frac{46 - 42}{3} = 1.3333 \dots$$

$$P(0.33 < z < 1.33) = P(z < 1.33) - P(z < 0.33)$$

$$= 0.90824 - 0.62930 \text{ (from the table)}$$

$$= 0.27894 \approx 0.279$$



- 6 In a class, the heights of students are normally distributed. The mean height of these students is 180 cm and the standard deviation is 7. Approximately what percentage of the class is shorter than 173 cm? 1

16%

- 7 The maximum temperature for each day has been recorded for Paris. The mean of these maximum temperatures during autumn is 22.5°C, their standard deviation is 3.7°C and the z-scores were calculated for every recorded temperature.

- a. What temperature has a z-score of -1 ? 1

18.8°

- b. What percentage of autumn days in Paris would have maximum temperatures between 3

11.4°C and 26.2°C? You may assume that these temperatures are normally distributed.

Temperature	11.4	15.1	18.8	22.5	26.2	29.9	33.6	$68\% \div 2 = 34\%$
z-score	-3	-2	-1	0	1	2	3	$99.7\% \div 2 = 49.85\%$
								$34\% + 49.85\% = 83.85\%$

- c. If the z-scores are normally distributed, what is the mean and standard deviation of all 2

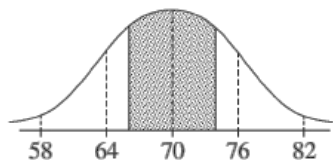
the z-scores? **Mean = 0, Standard deviation = 1**

- 8 The scores on an examination are normally distributed with a mean of 70 and a standard 1

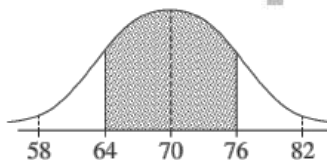
deviation of 6. Michael received a score on the examination between the lower quartile and the upper quartile of the scores. Which shaded region most accurately represents where Michael's score lies?

50% of candidate scores are between LQ and UQ.

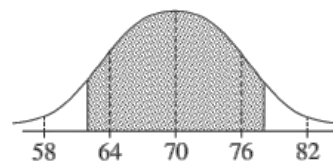
A.



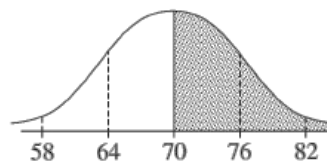
B.



C.



D.



- 9 In a particular country, the birth weight of babies is normally distributed with a mean of 3000 2

grams. It is known that 95% of these babies have a birth weight between 1600 grams and 4400 grams. One of these babies has a birth weight of 3497 grams. What is the z-score of this baby's birth weight?

$$\mu = 3000, \sigma = \frac{3000 - 1600}{2} = 700, z = \frac{3497 - 3000}{700} = 0.71$$

- 10 The Intelligence Quotient (IQ) scores for adults in City A are normally distributed with a mean of 108 and a standard deviation of 10. The IQ scores for adults in City B are normally distributed with a mean of 112 and a standard deviation of 16.

- a. Yin is an adult who lives in City A and has an IQ score of 128. What percentage of the adults in this city have an IQ score higher than Yin's?

1

$$A: P(IQ > 128) = P(Z > 2) \approx \frac{1 - 0.95}{2} \times 100\% = 2.5\%$$

- b. There are 1 000 000 adults living in City B.

Calculate the number of adults in City B that would be expected to have an IQ score lower than Yin's.

2

$$B: P(IQ < 128) = P(Z < 1) \approx 0.5 + \frac{0.68}{2} = 0.84$$

$$\text{Expected number} \approx 0.84 \times 1\,000\,000 = 840\,000$$

- 11 John recently did a class test in each of three subjects. The class scores on each test were normally distributed.

Subject	John's score	Mean	Standard deviation
French	82	70	8
Commerce	80	65	5
Music	74	50	12

1

The table shows the subjects and John's scores as well as the mean and standard deviation of the class scores on each test. Relative to the rest of the class, which row of the table below shows John's strongest subject and his weakest subject?

	Strongest subject	Weakest subject
A.	Commerce	French
B.	French	Music
C.	Music	French
D.	Commerce	Music

French $z = 1.5$;
 Commerce $z = 3$;
 Music $z = 2$

A